AI/ML intern DAY 12

Task — Virtual Try-On System Optimization

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Introduction

Day 12 focused on improving the performance, realism, scalability, and optimization of the Virtual Try-On (VTON) system for Raritone. The tasks included improving 2D try-on fitting accuracy, researching advanced 3D try-on systems, optimizing AI models, expanding datasets, and documenting the complete AI workflow pipeline.

The work concentrated on improving garment alignment, segmentation quality, pose estimation accuracy, rendering realism, inference performance, and dataset scalability using Artificial Intelligence, Machine Learning, and Computer Vision technologies.

The optimization workflow focused on:

- clothing alignmentP
- body segmentation
- pose estimation
- TPS garment warping
- rendering optimization
- AI inference acceleration
- dataset organization
- workflow documentation

Today's Improvement Goals

1. Improve 2D Virtual Try-On Accuracy

Focus Areas

- Better clothing alignment
- Improved body segmentation
- Accurate pose detection
- Realistic garment overlay
- Edge smoothing and fitting quality

Work Performed

- Human parsing optimization
- Clothing masking improvements
- Background handling optimization
- Different body pose testing
- TPS garment warping improvements
- Garment overlay refinement
- Edge smoothing enhancement

Current 2D Workflow

User Image + Garment Image

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Image Preprocessing

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Pose Detection

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Human Parsing

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Garment Warping

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Garment Overlay

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Rendering & Smoothing

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Final Try-On Output

Clothing Alignment Optimization

Existing Issues Identified

- Sleeve misalignment
- Shoulder mismatch
- Incorrect torso fitting

- Neck overlap
- Garment stretching artifacts

Optimization Methods Used

TPS (Thin Plate Spline) Warping

TPS warping was used to:

- deform garments according to posture
- improve sleeve fitting
- reduce deformation artifacts
- improve body-cloth alignment

DensePose Integration

DensePose improved:

- body surface understanding
- garment placement accuracy
- body-to-cloth mapping

OpenPose Landmark Guidance

Landmarks detected:

- shoulders
- elbows
- wrists
- hips
- torso points

These landmarks guided garment positioning and overlay alignment.

Human Parsing Optimization

Objective

Improve body segmentation quality and garment layering.

Problems Identified

- Arm segmentation errors
- Hair overlap
- Background leakage

- Torso masking issues

Technologies Explored

Detectron2

Used for:

- body separation
- background removal
- clothing mask generation
- segmentation refinement

SCHP Parsing

SCHP improved:

- human segmentation
- occlusion handling
- garment layering
- body region extraction

Segmented regions:

- face
- hair
- arms
- torso
- legs
- background

Pose Detection Optimization

Technologies Used

MediaPipe Pose

Advantages:

- fast inference
- low latency
- real-time tracking

Detected points:

- shoulder
- elbow
- wrist
- hip
- knee

OpenPose

Advantages:

- multi-point detection
- accurate torso fitting
- better pose robustness
- improved sleeve alignment

Pose Testing Scenarios

Test Case	Description
Pose 1	Front standing
Pose 2	Side orientation
Pose 3	Arms raised
Pose 4	Walking posture
Pose 5	Hand movement variations

Garment Overlay Optimization

Existing Problems

- floating garments
- colour mismatch
- layer inconsistency
- unrealistic fitting

Optimization Techniques

GAN-Based Rendering

GAN rendering improved:

- visual realism
- texture quality

- overlay consistency
- surface blending

Edge Smoothing Optimization

Techniques used:

- Alpha blending
- Feather masking
- Gaussian smoothing

Benefits:

- softer boundaries
- reduced artifacts
- realistic cloth blending

Background Handling Optimization

Problems Identified

- background leakage
- object interference
- segmentation noise

Portrait Matting

Portrait matting improved:

- background isolation
- body extraction
- segmentation quality
- overlay accuracy

Background Testing Cases

Scenario	Evaluation
Plain Background	Tested
Indoor Scene	Tested
Outdoor Scene	Tested
Complex Background	Tested

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2. Improve 3D Try-On Research & Prototype

Research Areas

- 3D avatar generation
- body mesh fitting
- cloth simulation
- movement tracking
- rendering optimization
- user measurement integration

Proposed 3D Workflow

User Image / Camera Input

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Body Detection

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Avatar Generation

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3D Mesh Creation

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Garment Fitting

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Cloth Simulation

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Motion Tracking

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Rendering Engine

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Final 3D Output

3D Avatar Generation

SMPL Model

Used for:

- parametric body meshes
- adjustable body proportions
- realistic avatar structures
- garment fitting support

PIFuHD

Capabilities:

- high-resolution reconstruction
- surface estimation
- detailed avatar generation

Cloth Simulation Research

PhysX Engine

Features:

- cloth physics
- collision handling
- dynamic movement
- realistic cloth response

Unreal Engine Cloth System

Capabilities:

- dynamic cloth simulation
- real-time rendering
- realistic visualization
- physics integration

Avatar Movement Tracking

MediaPipe Integration

Used for:

- body landmark extraction

- movement synchronization
- avatar motion mapping

Detected points:

- shoulder
- elbow
- wrist
- hip
- knee
- ankle

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3. AI Model Optimization

Objectives

- Improve landmark detection accuracy
- Reduce processing time
- Increase inference speed
- Optimize segmentation quality
- Improve recommendation systems
- Reduce latency

Landmark Detection Optimization

OpenPose Optimization

Advantages:

- high precision
- multi-point tracking
- accurate fitting

Applications:

- pose estimation
- body mapping
- garment alignment

MediaPipe Optimization

Benefits:

- fast execution
- lightweight inference
- mobile compatibility
- low computation

Lightweight AI Models

MobileNet

Advantages:

- lightweight architecture
- faster inference
- edge deployment support

EfficientNet

Benefits:

- balanced performance
- lower computational cost
- improved accuracy

Inference Optimization

ONNX Optimization

Benefits:

- faster execution
- reduced overhead
- cross-platform deployment

TensorRT Acceleration

Improved:

- GPU inference
- prediction speed
- memory efficiency
- latency reduction

Recommendation System Optimization

Inputs Used

- clothing category
- body type
- size information
- user preferences

Recommendation Flow

User Input

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Body Analysis

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Clothing Features

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Recommendation Engine

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Suggested Outfit

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4. Dataset Expansion & Cleaning

Objectives

- collect more fashion datasets
- improve clothing diversity
- add multiple body types
- increase pose variations
- improve dataset organization
- remove noisy samples

Fashion Datasets Used

Dataset	Use Case
DeepFashion2	Clothing analysis

VITON	Virtual Try-On
DressCode	Fashion diversity
Fashion-MNIST	Clothing classification

Clothing Categories Added

- shirts
- t-shirts
- dresses
- jackets
- tops
- casual wear
- formal wear
- traditional wear

Pose Variations Collected

Pose Type	**Description**
Front Pose	Standing
Side Pose	Slight rotation
Arms Raised	Hands above shoulder
Walking Pose	Dynamic movement
Bent Arm Pose	Hand movement

Structured Dataset Organization

```
dataset/
  users/
    male/
    female/

  garments/
    shirts/
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tops/

dresses/

jackets/

poses/

front/

side/

walking/

body_types/

slim/

medium/

heavy/

annotations/

Dataset Cleaning Process

Removed Samples

- duplicate images
- blurry samples
- corrupted files
- incorrect labels
- low-resolution images

Cleaning Workflow

Raw Dataset

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Duplicate Detection

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Blur Detection

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Label Verification

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Clean Dataset

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5. AI Workflow Documentation

Objectives

- document AI pipeline structure
- define input/output flow
- describe AI modules
- analyze dependencies
- identify challenges
- propose improvements

Overall AI Workflow

User Upload

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Image Preprocessing

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Pose Detection

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Human Parsing

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Garment Warping

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Rendering

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Final Try-On Output

AI Modules Covered

Preprocessing Module

Operations:

- resizing
- normalization
- background handling
- contrast adjustment

Pose Detection Module

Used for:

- body landmark extraction
- body orientation analysis
- garment alignment guidance

Human Parsing Module

Used for:

- segmentation
- mask generation
- body extraction

Garment Warping Module

Used for:

- TPS deformation
- body adaptation
- cloth alignment

Rendering Module

Used for:

- texture refinement
- realism enhancement
- artifact reduction

Proposed Deployment Architecture

Frontend

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Flask API

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Preprocessing Server

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AI Models

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GPU Inference

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Rendering Engine

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Final Output

Challenges Encountered

Clothing Related

- sleeve mismatch
- cloth deformation errors
- loose garment fitting

Human Parsing Related

- arm overlap
- hair occlusion
- incorrect masks

Pose Related

- side pose detection
- landmark shifts
- dynamic movement tracking

Environment Related

- lighting variation
- complex backgrounds
- segmentation noise

System Challenges

- GPU dependency

- memory bottlenecks
- inference latency
- scalability

Future Improvements

Planned improvements include:

- HR-VITON integration
- better DensePose fitting
- advanced cloth simulation
- real-time optimization
- lightweight segmentation models
- improved recommendation logic
- multi-angle fitting
- full body tracking
- 3D integration
- edge deployment optimization

Deliverables

- Improved demo screenshots
- Optimization documentation
- AI workflow documentation
- Updated datasets
- Progress summary
- AI pipeline reports
- Segmentation research
- VTON optimization analysis

Technologies Used

- Python
- OpenCV

- MediaPipe
- OpenPose
- Detectron2
- TensorFlow
- PyTorch
- ONNX
- TensorRT
- Flask API
- Artificial Intelligence
- Machine Learning
- Computer Vision
- Deep Learning

Learning Outcomes

During Day 12 tasks, the following concepts and skills were learned:

- TPS garment warping
- DensePose alignment
- Human parsing optimization
- Pose estimation workflows
- AI inference acceleration
- Lightweight AI deployment
- Dataset organization
- Segmentation optimization
- Recommendation systems
- 3D avatar generation
- Cloth simulation
- AI workflow architecture
- GPU inference optimization
- Virtual Try-On systems

Conclusion

Day 12 focused on improving the scalability, realism, and performance of the Virtual Try-On system using Artificial Intelligence, Machine Learning, Computer Vision, TPS garment warping, segmentation optimization, AI inference acceleration, and workflow documentation.

The tasks improved understanding of:

- 2D Virtual Try-On optimization
- 3D avatar systems
- AI segmentation workflows
- TPS garment fitting
- AI model optimization
- recommendation systems
- dataset preparation
- rendering optimization
- scalable AI architectures
- real-time computer vision systems

This internship task focused on Artificial Intelligence, Computer Vision, Human Parsing, Virtual Try-On Systems, AI Optimization, TPS Warping, Dataset Engineering, Pose Estimation, Cloth Simulation, and AI Workflow Architecture.